

Miscellaneous

1. The alphabet X in X-ray stands for _____ ray.

- (a) express (b) unknown
- (c) infrared (d) xenon
- (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (b)

Röntgen referred to the radiation as "X", to indicate that it was an unknown type of radiation. X-radiation is a form of electromagnetic radiation. They are powerful waves of electromagnetic energy. Most of them have a wavelength ranging from 0.01 to 10 nanometres, corresponding to frequencies in the range 3×10^{16} Hz to 3×10^{19} Hz and energies in the range 100 eV to 100 keV.

2. Water gas is a mixture of

- (a) CO + H₂ (b) CO + O₂
- (c) CO + N₂ (d) CO₂ + H₂
- (e) None of the above / More than one of the above

B.P.S.C. CDPO 2022

Ans. (a)

Water gas is a mixture of carbon monoxide as well as hydrogen produced from synthesis gas. It is a fuel gas in which carbon monoxide and hydrogen are present in equal proportions along with smaller amounts of other gases such as carbon dioxide, methane as well as Nitrogen.

3. Bithional is generally added to the soaps as an additive to function as a/an

- (a) softener (b) dryer
- (c) buffering agent (d) antiseptic
- (e) None of the above / More than one of the above

B.P.S.C. Assistant Audit Officer 2022

Ans. (a)

Bithional is added to toilet soaps to reduce odour produced by the bacterial decomposition of organic matter. This is possible because it is an antiseptic which is harmful to biological organisms like bacteria.

4. A solution turns red litmus blue, its pH is likely to be

- (a) 10 (b) 1
- (c) 4 (d) 5

B.P.S.C. Assistant Mains (GK) Paper II 2023

Ans. (a)

A red litmus turns blue when it comes in contact to any base and blue litmus turns red when it comes in contact to any acid. As we know pH value less than 7 shows acid and above pH value 7 shows base. Hence from the given options, its pH is likely to be 10.

5. You have been provided with three test tubes. One of them contains distilled water and the other two contain an acidic solution and a basic solution, respectively. Which of them will turn red litmus to blue?

- (a) Distilled water (b) Acid
- (c) Base (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

An alkaline or basic substance turns red litmus paper to blue. Litmus paper is a commonly used pH indicator. When basic substance applied to red litmus paper, the hydroxide ion reacts with the red dye in the litmus paper.

On the pH scale, a pH value greater than 7 indicates a basic or alkaline solution substances with a pH value above 7 has a higher concentration of hydroxide ions (OH⁻)

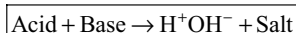
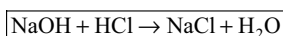
6. Sodium hydroxide and hydrochloric acid react to form sodium chloride and water. Which reaction is this?

- (a) Neutralization (b) Dissolution
- (c) Combination (d) More than one of the above
- (e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (a)

The reaction of an acid and a base to produce salt and water is known as neutralization reaction. Thus when sodium hydroxide and hydrochloric acid react to form sodium chloride and water is neutralization reaction.



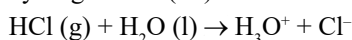
7. Why does dry HCl gas not change the colour of dry litmus paper?

- (a) HCl gas acts as a dehydrating agent.
- (b) Blue litmus becomes dry in the presence of dry HCl gas.
- (c) No H₃O⁺ ions are present. Litmus changes colour only in the presence of H₃O⁺ ions.
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

Dry HCl gas does not change the colour of dry litmus paper because HCl gas is non-polar and does not produce any hydrogen ions (H⁺) in the absence of water.



In this reaction, hydrogen ion (H₃O⁺) are released, making the solution acidic, and litmus paper would turn red. But in absence of water, the dry HCl gas does not lead to any colour change in dry litmus paper.

8. Some metal oxides react with acid as well as base to produce salt and water. They are known as amphoteric oxides. The examples of amphoteric oxides are

- (a) CuO and ZnO (b) Al_2O_3 and ZnO
(c) Al_2O_3 and CuO (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

Metal oxides that can react with acids and bases to produce a salt and water are called amphoteric oxides. These oxides exhibit amphoterism, meaning they display both acidic and basic properties depending on the reaction conditions. Some examples are—Aluminum Oxide (Al_2O_3), Zinc Oxide (ZnO), Lead Oxide (PbO), Tin Oxide (SnO_2) etc.

9. Detect the odd one.

- (a) Ripening of fruit (b) Setting of cement
(c) Burning of coal (d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (e)

All the given option are the examples of chemical changes. That's why option (e) will be correct. Chemical changes, are processes in which substances undergo a transformation, resulting in the formation of new substances with different properties.

10. Why gelatin is used a lot in making ice-cream?

- (a) To prevent colloid formation
(b) In stabilizing colloid and preventing crystal formation
(c) To stabilize the mixture
(d) To make it scented
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

Gelatin acts as an emulsifier. It is used to make the ice cream to get a soft texture and fresh appearance. As it is used in stabilizing the colloid and preventing crystal formation.

11. Which of the following is not a colloid?

- (a) Milk (b) Blood
(c) Ghee (d) Butter
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (e)

In the given options all the options are example of colloidal solution so option (e) is correct.

12. On mixing two colorless gases, dark brown gas is formed. These gases are

- (a) NO_2 and O_2 (b) N_2O and NO
(c) NO and O_2 (d) NH_3 and HCl
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (c)

The chemical name of the deep coloured brown gas is nitrogen dioxide and its chemical formula is NO_2 . Both NO and O_2 gases are colourless and when nitrogen monoxide reacts with dioxygen in warm conditions, then the formation of deep coloured brown gas i.e., nitrogen dioxide takes place.

13. Which compound is known as 'oil of wintergreen'?

- (a) Phenyl benzoate (b) Phenyl salicylate
(c) Phenyl acetate (d) Methyl salicylate
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (d)

Methyl salicylate is an organic compound with the formula $\text{C}_8\text{H}_8\text{O}_3$. It is the methyl ester of salicylic acid. It is known as oil of wintergreen.

14. The Nobel Prize in Chemistry, 2019 was awarded for the development of :

- (a) LED (b) LCD
(c) Lithium-ion batteries (d) MRI
(e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (c)

The Nobel Prize in Chemistry, 2019 was awarded to John B. Goodenough (USA), M. Stanley Whittingham (USA) and Akira Yoshino (Japan) for the development of lithium-ion batteries. The Nobel Prize in chemistry, 2020 was awarded to Emmanuelle Charpentier (Germany) and Jennifer A. Doudna (USA) for the development of a method for genome editing (CRISPR/Cas 9 genetic scissors). The Nobel Prize in Chemistry 2021 was awarded jointly to Benjamin List (Germany) and David W.C. MacMillan (Scotland, UK) "for the development of asymmetric organocatalysis".

15. An active chemical to remove mosquitoes is-

- (a) Allethrin (b) Atropin
(c) 2-ISO propoxiphinyal (d) BenzeneHexachlorophene

41st B.P.S.C. (Pre) 1996

Ans. (a)

Allethrin is a pyrethroid synthesized insecticide/pesticide. It is an active chemical to remove mosquitoes.

16. Gamaxine insecticide is based on

- (a) DDT (b) benzene hexachloride
(c) hexachlorobenzene (d) chloral
(e) None of the above / More than one of the above

B.P.S.C. Auditor 2021

Ans. (b)

Gammexane is commonly known as Benzene hexachloride (BHC) which is used as insecticide on crops, in forestry, for seed treatment. It is used in the treatment of head and body lice. It is used in pharmaceuticals. It is used to treat scabies.

17. Which chemical substance is used for making rat poison?

- (a) Ethyl Alcohol (b) Methyl Isocyanate
(c) Potassium Cyanide (d) Ethyl Isocyanide

47th B.P.S.C. (Pre) 2005

Ans. (c)

Potassium cyanide (KCN) and zinc phosphide which are highly toxic chemicals, used as a rodenticide - a poison to kill the mouse/rat.

18. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I	List-II
A. Skin Cancer	1. Chlorofluorocarbons
B. Noise Pollution	2. Ultraviolet light
C. Global Warming	3. Decibel
D. Ozone Hole	4. Carbon dioxide

Code :

- | | | | |
|-------|---|---|---|
| A | B | C | D |
| (a) 1 | 2 | 3 | 4 |
| (b) 2 | 3 | 4 | 1 |
| (c) 2 | 3 | 1 | 4 |
| (d) 4 | 3 | 2 | 1 |

- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

The correct matching of List-I and List-II is as follows –

Skin Cancer	-	Ultraviolet Light
Noise Pollution	-	Decibel
Global Warming	-	Carbon dioxide
Ozone Hole	-	Chlorofluorocarbon

19. SO₂ pollution is indicated by

- (a) orchids (b) mangroves
(c) lichens (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (c)

The largest source of SO₂ in the atmosphere is the burning of fossil fuels by power plants and other industrial facilities. SO₂ can affect both health and environment. Short-term of exposures to SO₂ can harm the human respiratory system and make breathing difficult. People with asthma, particularly children, are sensitive to these effects of SO₂. This Pollution is indicated by lichens, if air is very badly polluted with sulfur dioxide there may be no lichens present, just green algae may be found.

20. Which of the following gases is produced from landfills?

- (a) Sulphur dioxide
(b) Methane
(c) Liquefied petroleum gas
(d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift I

Ans. (b)

Methane and carbon dioxide make up 90 to 98% of landfill gas. The remaining 2 to 10% includes nitrogen, oxygen, ammonia, sulfides, hydrogen and various other gases. Landfill gases are produced when bacteria break down organic waste.

21. In a common dry cell, electrolyte is –

- (a) Zinc (b) Sulphuric acid
(c) Ammonium chloride (d) Manganese dioxide

39th B.P.S.C. (Pre) 1994

Ans. (c)

A dry cell is a type of electrochemical cell, commonly used in the form of a battery in many electrical appliances, torch, calculator etc. Unlike a wet cell, a dry cell can operate in many orientations without spilling as it contains no free liquid. A common dry cell is a zinc-carbon battery, sometimes it is called the dry Leclanche cell. Ammonium chloride is used as the electrolyte. Sometimes ammonium chloride is replaced by zinc chloride for having high energy voltage.

22. In a car battery, electrolyte substance used is-

- (a) Hydrochloric acid (b) Sulphuric acid
(c) Nitric acid (d) Distilled water

42nd B.P.S.C. (Pre) 1997

Ans. (b)

Electrolyte substance used in a car battery is sulphuric acid (H₂SO₄). It is a strong acid.

23. The acid used in a car battery is :

- (a) Acetic acid (b) Hydrochloric acid

- (c) Nitric acid (d) Sulfuric acid
(e) None of the above/More than one of the above

63rd B.P.S.C. (Pre) 2017

Ans. (d)

See the explanation of the above question.

24. Match List-I with List-II and select the correct answer using the code given below the lists :

List-I (Substances)	List-II (Applications)
A. Aspartame	1. Synthetic rubber
B. Freon	2. Anti-histamine
C. Neoprene	3. Artificial Sweetener
D. Benadryl	4. Refrigerant

Code :

	A	B	C	D
(a)	1	3	4	2
(b)	2	4	1	3
(c)	3	4	1	2
(d)	3	1	2	4

- (e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (c)

The correct matching is as follows :

Aspartame	-	Artificial Sweetener
Freon	-	Refrigerant
Neoprene	-	Synthetic Rubber
Benadryl	-	Anti-histamine

25. The United Nations Framework Convention on Climate Change deals with

- (a) reduction in fossil fuel usage
(b) CO₂ emissions mitigation
(c) reduction in Uranium production
(d) greenhouse gases emissions mitigation
(e) None of the above/More than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (e)

The United Nations Framework Convention on Climate Change deals with greenhouse gases emission mitigation. It is known fact that CO₂ falls under greenhouse gases. On 22 April, 2016, 175 countries/organisations including India signed the historic "Paris Climate Change Agreement" to reduce carbon emission. The participating countries had agreed that the average temperature of 21st century should not be increased by more than 2°C in comparison to the temperature of pre-industrial era.

26. The incomplete burning of petrol and diesel produces :

- (a) Nitric oxide (b) Nitrogen dioxide
(c) Carbon dioxide (d) Carbon monoxide
(e) None of the above / More than one of the above

66th B.P.S.C. (Pre) 2020

Ans. (d)

In the complete combustion (burning) of hydrocarbon fuels (i.e. petrol and diesel), hydrocarbons react with oxygen and water and carbon dioxide are produced. Incomplete combustion occurs when the supply of air or oxygen is poor. In the incomplete combustion of hydrocarbon fuels water is still produced, but carbon monoxide (a poisonous gas) and carbon (as soot) are produced instead of carbon dioxide.

27. What is particulate matter?

- (a) Solid residue (b) Air pollutant
(c) Water pollutant (d) Soil pollutant
(e) None of the above/more than one of the above

60th to 62nd B.P.S.C. (Pre) 2016

Ans. (b)

Particulate matter (PM) also known as particle pollution, is a complex mixture of extremely small particles and liquid droplets that get into the air. Once inhaled, these particles can affect the heart and lungs and cause serious health problems.

28. Which of the following is a secondary pollutant?

- (a) PAN (b) SO₂
(c) Ozone (d) More than one of the above
(e) None of the above

B.P.S.C. PRT 2023 Shift II

Ans. (d)

Primary pollutants are directly emitted from a source while secondary pollutants arise when primary pollutants reacts with one another or with other substances in the surrounding Peroxyacetyl Nitrate (PAN), SO₂ Ozone are secondary pollutants.

29. Which of the following gases is considered as atmospheric pollutant?

- (a) Argon (b) Oxygen
(c) Sulphur dioxide (d) More than one of the above
(e) None of the above

B.P.S.C. School Teacher (Class 6-8) 10-12-2023

Ans. (c)

From the given options, Sulphur dioxide is considered as atmospheric pollutant. Sulfur dioxide in the air comes mainly from activities such as the burning of coal and oil at power plants or from copper smelting. In nature, sulfur dioxide can be released into the air from volcanic eruptions.